



OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 7

Team No.: 26

Project Title: Database Client Server Management system

Team Members

Roll Number	Student Name	Signature
2520030594	P.Devi sree	
2520030127	K.Shresta priya	
2520030116	Hasini Reddy	

1. Abstract

The Database Client-Server Management System is a Linux-based application that demonstrates client-server communication and database management using Operating Systems and Systems Programming concepts. The system consists of a client that sends database-related requests to a server, while the server processes these requests and interacts with the database. Socket programming is used to establish communication between the client and server, enabling operations such as adding, viewing, updating, and deleting records.

The server is designed to handle multiple client requests and uses Linux system calls, process management, synchronization, and resource management mechanisms. The project demonstrates important concepts such as processes, inter-process communication, sockets, file handling, and client-server architecture.

The expected outcome is a functional Linux-based application that allows users to perform basic database operations through a client while the server efficiently manages requests and database interactions.

2. Problem Statement

Managing database operations through a centralized server requires proper communication between clients and the server. A system is needed to demonstrate how multiple clients can send requests to a server and how the server processes these requests while maintaining reliable communication and database access.

The project addresses this problem by developing a Database Client-Server Management System in Linux. It focuses on applying Operating Systems and Systems Programming concepts such as process creation, socket communication, inter-process communication, synchronization, and system calls. The project provides practical understanding of how a server receives client requests, processes database operations, and returns responses to clients.

3. Objectives

1. To develop a Linux-based client-server application for performing basic database operations.
2. To implement communication between the client and server using socket programming.
3. To demonstrate Operating Systems concepts such as process creation, synchronization, inter-process communication, and system calls.
4. To provide a simple and reliable system for adding, viewing, updating, and deleting database records through the client.

4. Proposed Methodology

The system will be developed in Linux/Ubuntu using a client-server architecture. The client sends database requests through sockets, and the server processes operations such as INSERT, SELECT, UPDATE, and DELETE. Linux system calls and POSIX APIs will be used for socket communication, process management, file handling, and synchronization.

Client → Server → Database Operation → Server Response → Client.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
socket()	Creates a communication socket between client and server.
bind()	Assigns an IP address and port to the server socket.
listen()	Makes the server ready to accept client connections.
accept()	Accepts incoming client connections.
connect()	Establishes a connection from client to server.
send() / recv()	Sends requests and receives responses between client and server.
fork()	Creates a child process to handle client requests.
wait() / waitpid()	Synchronizes parent and child processes.
open() / read() / write() / close()	Demonstrates Linux file handling and system-level I/O.
IPC	Supports communication and coordination between processes.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030594	P.Devi sree	Client-side development, user input handling, and socket communication.
2520030116	Hasini Reddy	Database operations, file handling, synchronization, testing, and system integration.
2520030127	K.Shresta priya	Server-side development, client request handling, and process management using fork().

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and executing the project.
C / C++	Implementation of client-server and system programming components.
GCC Compiler	Compiling the C/C++ source code.
Socket Programming	Communication between client and server.
MySQL	Storing and managing database records
VS Code	Writing and managing source code.
Git / GitHub	Version control and project repository management.
Linux Terminal	Compiling, executing, and testing the application.

8. Expected Outcome

The project will produce a Linux-based Database Client-Server Management System. The client can connect to the server and perform operations such as adding, viewing, updating, and deleting records.

The server will process requests, interact with the database, and send responses to the client. The project will demonstrate socket communication, process management, synchronization, inter-process communication, and Linux system calls.

9. GitHub Repository

GitHub Repository Link: https://github.com/kallushrestapriya-byte/OS_Database-Client-Server-Management-System

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____